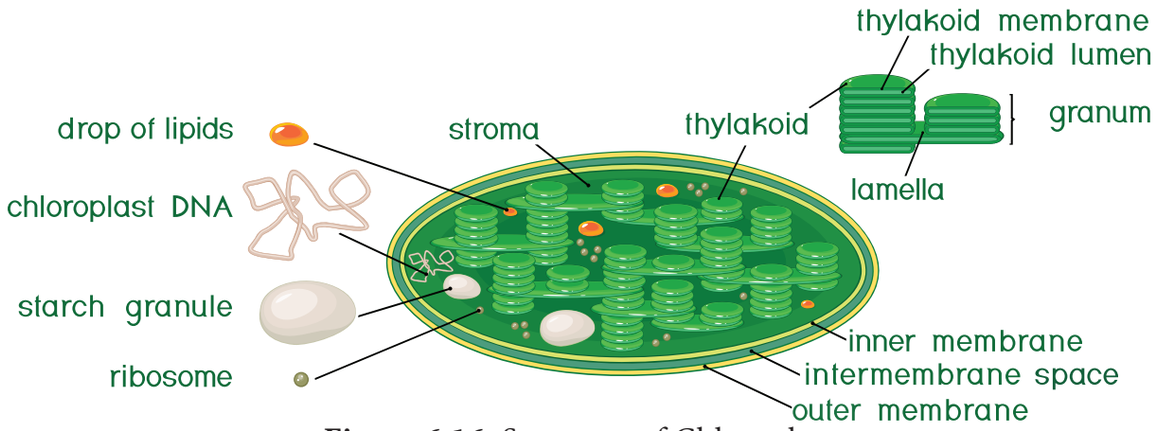


Figure 6.16: Structure of Chloroplast

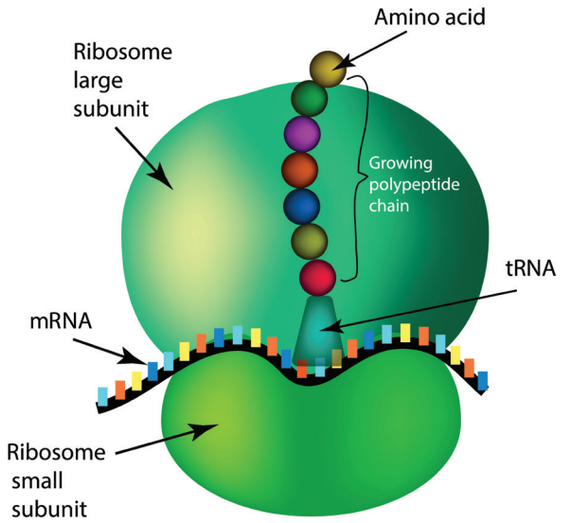


Figure 6.17: Structure of Ribosomes



Svedberg unit (s).

The size of ribosomes and their subunits are usually given in Svedberg unit (named after Theoder Svedberg, Swedish Chemist Noble Laureate 1929), a measure of a particle size dependent on the speed with which particle sediment in the ultracentrifuge.

Ribosome consists of RNA and protein: RNA 60 % and protein 40%. During protein synthesis, many ribosomes are attached to the single mRNA and is called **polysomes** or **polyribosomes**. The function of polysomes is the formation of several copies of a particular polypeptide during protein synthesis. They are free in non-protein synthesising cells. In protein synthesising cells they are linked together with the help of Mg^{2+} ions.

Types of Ribosomes

70S Ribosomes (sub unit 30S and 50S)	80S Ribosomes (sub units 40S and 60S)
3 RNA molecule (i) 16SrRNA in 30S subunit (ii) 23S and 5S in 50S large subunit	4 RNA molecule (i) 18SrRNA in 40S small subunit (ii) 28S, 5.8S and 5S in larger 60S subunit
(Prokaryotic cells of bluegreen algae bacteria, mitochondria and chloroplast of many algae and higher plants)	**(Eukaryotic cells of plants and animals)**

6.6.8 Lysosomes (Suicidal Bags of Cell)

Lysosomes were discovered by **Christian de Duve** (1953), these are known as **suicidal bags**. They are spherical bodies enclosed by a single unit membrane. They are found in eukaryotic cell. Lysosomes are small vacuoles formed when small pieces of golgi body are pinched off from its tubules.

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